Closing the case of two opposite six-point heads

Human Compression project: L044 continuation

Retain the complete critical-head hypotheses and their acquired proofs, including the smaller-degree alternating induction that produced this configuration. Thus $G \leq \mathrm{Alt}(\Omega)$ is transitive with a connected string C-marking $\rho_0, \ldots, \rho_{r-1}$, all maximal omissions are intransitive, and $n = 2r \geq 18$. Suppose the first and last three marks are actual supported, extendable six-point Table 2 heads. Write their disjoint supports as D_L, D_R, with exit vertices v_L, v_R. The complete critical-head theorem and its full reversal give

$$G_{\geq 3} = \mathrm{Alt}(\Omega \setminus (D_L \setminus \{v_L\})), \qquad C := G_{\leq r-4} = \mathrm{Alt}(V), \quad V = \Omega \setminus (D_R \setminus \{v_R\}). \tag{1}$$

All groups in (1) are actual natural permutation subgroups. On the left use the literal notation

$$\tau = \rho_3 = (v_L\ \beta)\delta, \quad \sigma = \rho_4, \quad H_C = \langle \rho_5, \ldots, \rho_{r-4} \rangle, \quad K_C = \langle \rho_6, \ldots, \rho_{r-4} \rangle.$$

The odd involution δ fixes D_L, β and centralizes $G_{\geq 5}$. The first three marks are supported on D_L; σ fixes D_L; $G_{\geq 5}$ fixes $D_L \cup \{\beta\}$. Every use of eligibility below refers to the full group specified.

Lemma 1 (The central colours are paired)**.** *Every colour $3, \ldots, r-4$ has two edges eligible for its full ambient omission in G.*

Proof. The complete transitive-B exclusion and early-cut calculation give paired colours $3, 4$, and their full reversals give paired colours $r-4, r-5$. It remains to consider $5 \leq i \leq r-6$. If colour i had just one eligible edge, the acquired pure-prefix orbit lemma would make every mark after i fix the omission orbit containing D_L pointwise. Apply the entire lemma in reversed order: every mark before i fixes the other omission orbit, containing D_R, pointwise. These are the two omission orbits, so the cut has actual supported transitive prefix and suffix and is unshared.

The complete later-unshared-cut theorem applies to this literal cut. Its only residual indices are $i = r-3$ and $i = r-1$, both outside the present range. Thus no such unique cut exists. There is at least one eligible edge at every colour, because the full group is transitive and its omission is intransitive. Consequently every colour in the range is paired. □

Lemma 2 (An odd prefix and the actual attachment)**.** *The complete odd-degree separator arguments apply to C on V. They give an index s with $5 \leq s < r-4$ such that*

$$|\operatorname{Fix}_\Omega(G_{>s})| \geq 2s + 1. \tag{2}$$

Their local fixed points survive in this full ambient fixed set.

Proof. The degree and rank of the actual marking of C are $|V| = n - 5 = 2(r-3) + 1$ and $p = r - 3 \geq 6$. Restriction to V is faithful, so actual IP, rank and connectedness are retained. If a maximal omission C_i were transitive on V, adjoining the right head, whose support meets V at exactly v_R, would make G_i transitive on Ω. Hence all its maximal omissions are intransitive.

Intersect the two actual natural groups in (1). Full IP gives

$$C_{\geq 3} = \langle \rho_3, \ldots, \rho_{r-4} \rangle = \mathrm{Alt}(V \setminus (D_L \setminus \{v_L\})).$$

The literal left head and its odd δ are unchanged: every mark of C fixes $\Omega \setminus V$, and the core transposition is inside V. The previous lemma supplies two ambient G_i-eligible edges for every

$5 \le i \le r-4$. These edges lie in V and are C_i-eligible because $C_i \le G_i$. This is precisely the valid direction of eligibility inheritance. All hypotheses of the complete odd separator are now paid.

It yields $s < p-1 = r-4$ and at least $2s+1$ points of V fixed by $\langle \rho_{s+1}, \dots, \rho_{r-4} \rangle$. The right exit vertex v_R is not among them: the literal right crossing ρ_{r-4} moves it. Each of the three remaining marks is supported on D_R and hence fixes $V \setminus \{v_R\}$ pointwise. Thus all these local fixed points are fixed by the full $G_{>s}$, proving (2). □

Lemma 3 (The exterior involution fixes the later support)**.** *For the separator s just obtained, put $R = G_{>s}$, $U = \mathrm{Supp}(R)$, $F = \Omega \setminus U$, and $d = r-s-1$. Then*

$$|F| \le 2s+2, \qquad \delta|_U = 1. \tag{3}$$

Proof. Here $d \ge 4$, and the full reversed pure-prefix theorem applies to R. Its orbit A containing D_R has at least seven points and carries the pure natural $\mathrm{Alt}(A)$. Let $q \ge 1$ be the number of nontrivial R-orbits. The actual marked intersection

$$J = R \cap C = \langle \rho_{s+1}, \dots, \rho_{r-4} \rangle$$

has rank $d-3$. The exact reversed prefix intersection formula removes just the five points $D_R \setminus \{v_R\}$ from the head orbit and its support, leaving every exterior orbit unchanged. Thus its support has size $|U|-5$ and it has q nontrivial orbits. The reduced head orbit $A' = A \cap V$ is nontrivial even when $|A| = 7$: its two points are exchanged by the literal ρ_{r-4}. For larger A its pure alternating subgroup is transitive on A'.

All $d-3$ colours of J are paired by Lemma 1. The full ambient interval inequality therefore gives

$$2(d-3) \le |U| - 5 - q, \qquad |U| \ge 2d - 1 + q. \tag{4}$$

On the other hand (2) gives $|U| \le n - 2s - 1 = 2d+1$. Consequently $q \le 2$, and (4) already proves $|F| \le 2s+2$.

The involution δ centralizes R and hence its pure $\mathrm{Alt}(A)$, so fixes A pointwise. If $q = 1$, this proves the second assertion of (3). Suppose $q = 2$, and let E be the other nontrivial orbit. Equality in (4) is forced:

$$|U| = 2d+1. \tag{5}$$

Centralization makes δ preserve E. If it were nontrivial there, it would be fixed-point-free, so E would consist of $|E|/2$ pairs of δ.

Partition the colours of J into L, those with an ambient-eligible edge in E, and B, the remaining colours. Choose one such edge for each colour of L and project it to the pair set. Its endpoints are distinct pairs: the actual element $\tau \in G_i$ induces δ on E. These projected edges form a forest. An alternate path avoiding colour i would lift using the other selected generators and τ, all belonging to G_i, and would connect the eligible edge's endpoints inside one G_i-orbit. Thus $|L| \le |E|/2 - 1$.

For every colour of B, both eligible edges are confined to A'. The complete selected-localization theorem for arbitrary subsets of the interval of J applies to this single actual J-orbit, giving $2|B| \le |A'| - 1 = |A| - 6$. Therefore (5) gives

$$d-3 = |L| + |B| \le \frac{|E|}{2} - 1 + \frac{|A|-6}{2} = \frac{2d+1-8}{2} = d - \frac{7}{2},$$

a contradiction. So δ fixes E as well, proving (3). This is the acquired exterior-deficit mechanism with its actual omitted generators and localized orbit specified. □

Theorem 4. *The two opposite actual six-point heads cannot occur in the critical configuration.*

Proof. We retain the actual witness construction of the odd separator inside C, rather than infer an additional property from its numerical statement. First suppose it uses two witnesses intransitive under K_C. All their points are fixed by δ. If their endpoints are equal to s, the seven left head

points and the two whole orbits supply at least $2s+1$ δ-fixed points in the local fixed set. If their endpoints are unequal and the smaller endpoint s is at least $j+1$, the whole shorter orbit and the proved truncated longer orbit supply at least $2s$ such points, again including the seven head points.

For the remaining crossing-endpoint case the shorter orbit ends at j. The separator at $s=j+1$ uses its whole orbit and the two left endpoint orbits in the longer witness. Their respective sizes are at least $j-3$ and $j-3$, so with the seven head points they give $2j+1=2s-1$ δ-fixed points. This includes $j=5, s=6$, where both sizes are exactly two. If that exceptional case instead chooses the longer endpoint $s=y$, its proof gives at least $2y+2$ points fixed by both δ and the later tail, including the actual transported orbit supplied there. These are all the intransitive-witness cases. Lemma 2 transfers each displayed set to F.

Thus at least $2s-1$ points of F are δ-fixed. By (3), δ fixes U and at most three points of F remain available. Since δ is an odd involution, it is a literal transposition. This is excluded by the complete exterior-pair obstruction inside C: it has odd degree $2p+1$, its actual natural alternating tail and left head are unchanged, and every required later colour is paired in its full local ambient sense, as checked in Lemma 2.

It remains that the separator uses a K_C-transitive witness O. Its actual transport gives a disjoint K_C-orbit T in a K_C-intransitive witness. The complete proof gives

$$s \geq 6, \qquad |O| = |T| = N \geq s-4, \qquad D_L \cup \{\beta, \beta^\sigma\},\ O,\ T \subseteq F,$$

with these sets disjoint. The eight displayed head points and T are fixed pointwise by δ. If δ preserves O, its restriction there equals the actual $\tau \in C_j$. It preserves the unique ambient C_j-eligible witness edge and fixes its endpoints individually. Centralization of the transitive marked K_C-action then makes it fix all of O. At most $|F|-(8+2N) \leq 2$ points remain, so oddness again makes δ the already excluded literal transposition.

Otherwise O^δ is a distinct actual K_C-orbit in F, disjoint from O, T and the eight fixed points. It must fit in at most two remaining points. Thus $N=2$, $s=6$, and F consists exactly of the eight points and the three two-point orbits. The involution exchanges O, O^δ by two transpositions and fixes everything else in F; it also fixes U. It is even, a final contradiction. Every witness route is therefore impossible, proving the theorem. □

Attribution and remaining scope. The source is the opposite-end branch of Proposition 5.32 in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1. The complete critical heads, actual prefix kernels, paired intervals, odd separator proofs and exterior-pair obstruction remain acquired. The present argument proves the actual fixed-point transfer, combines one separator with the reversed prefix orbit deficit, and uses the odd exterior involution on the resulting fixed set. It requires neither an unproved ordering of two separators nor a new induction or finite classification. The terminal singleton attachment and full L044 source and induction assembly remain separate. No whole-original, question, numerical or formal credit is claimed here.